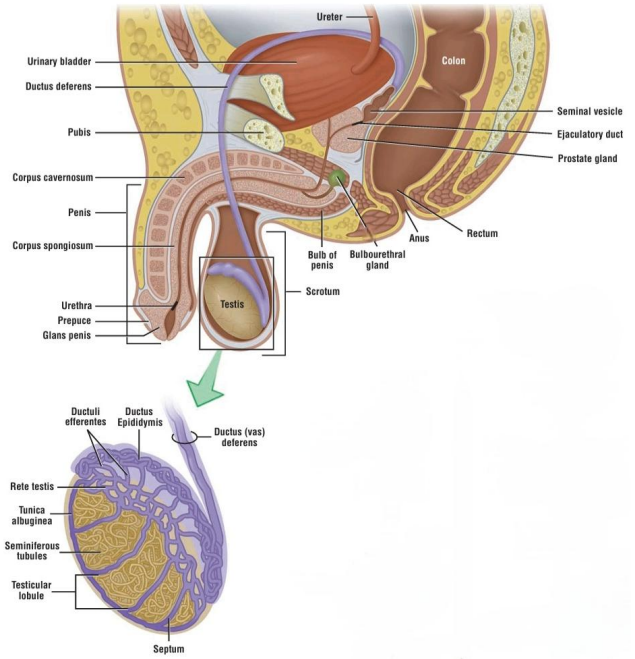
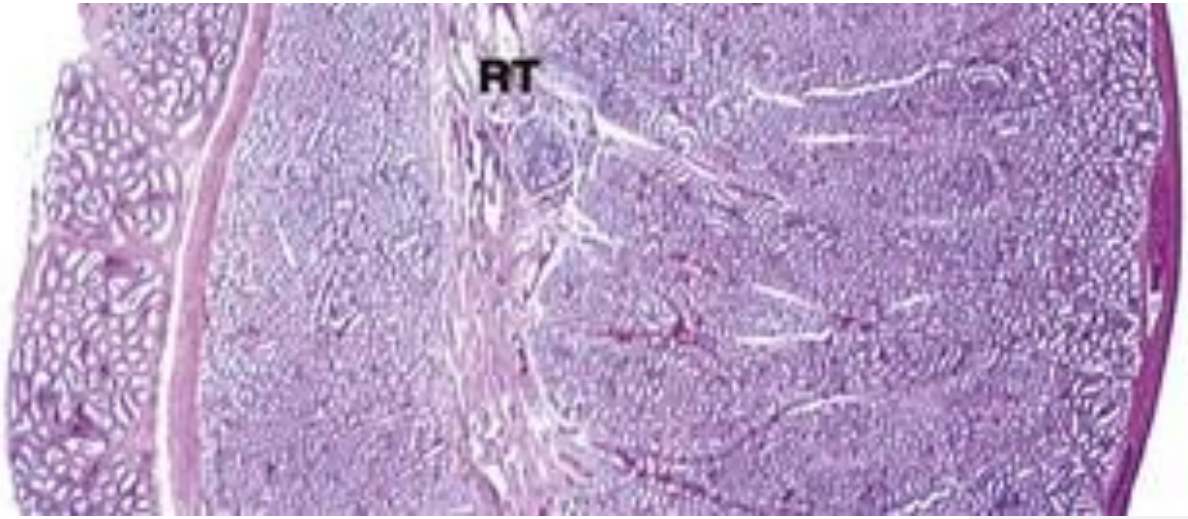


HISTOLOGY OF MALE REPRODUCTIVE SYSTEM

MANIPAL UNIVERSITY COLLEGE MALAYSIA

Formerly known as MELAKA-MANIPAL MEDICAL COLLEGE

Professor Dr Kalyan Chakravarthi Kosuri



#For instruction purpose only..

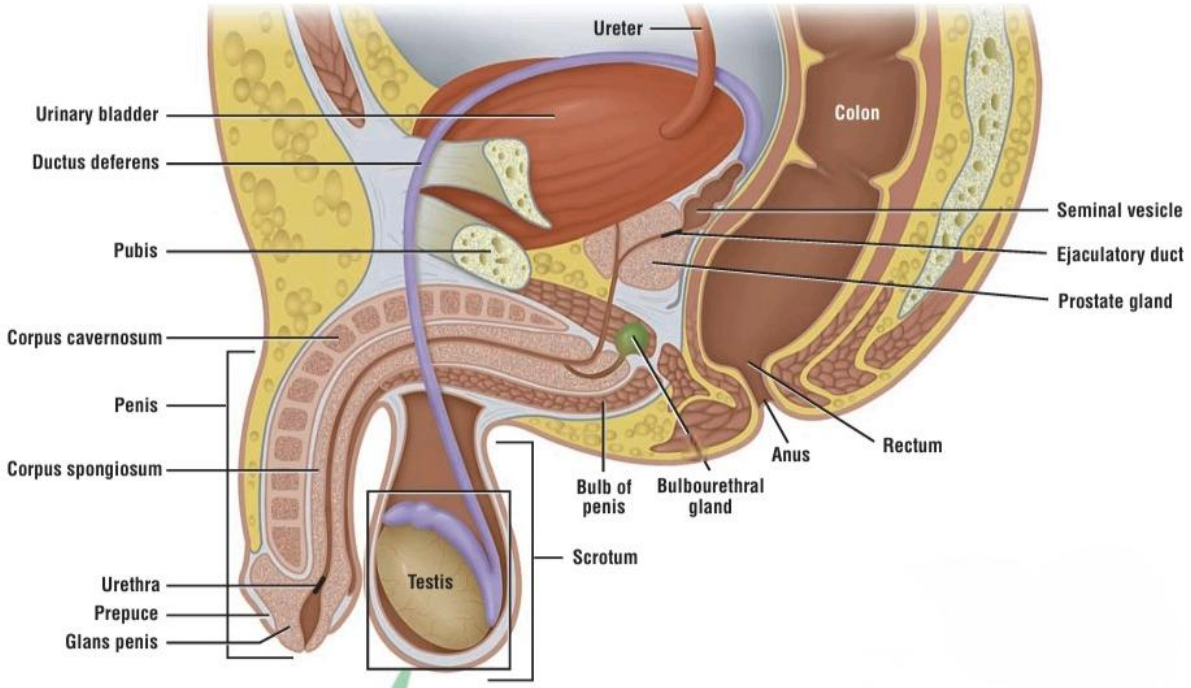
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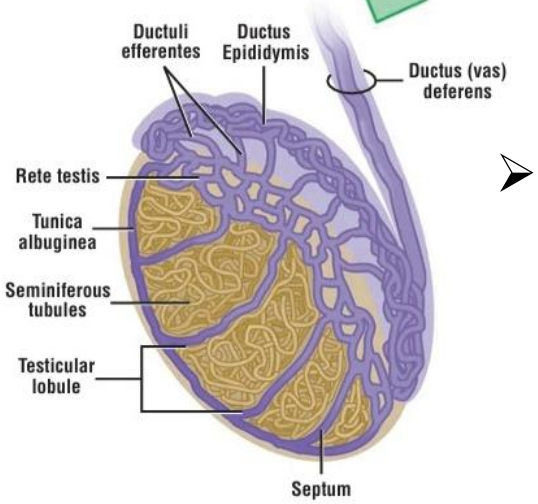
#inspire

LEARNING OBJECTIVES

1. Explain and illustrate the microscopic structure of the testis.
2. Explain and illustrate the microscopic structure of the prostate.

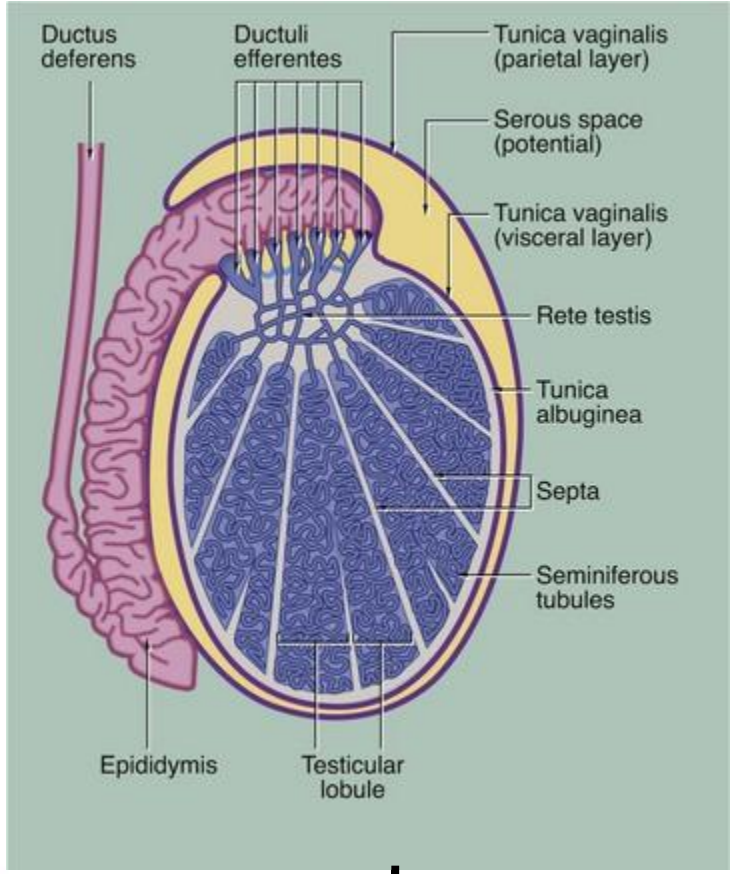
MALE REPRODUCTIVE SYSTEM



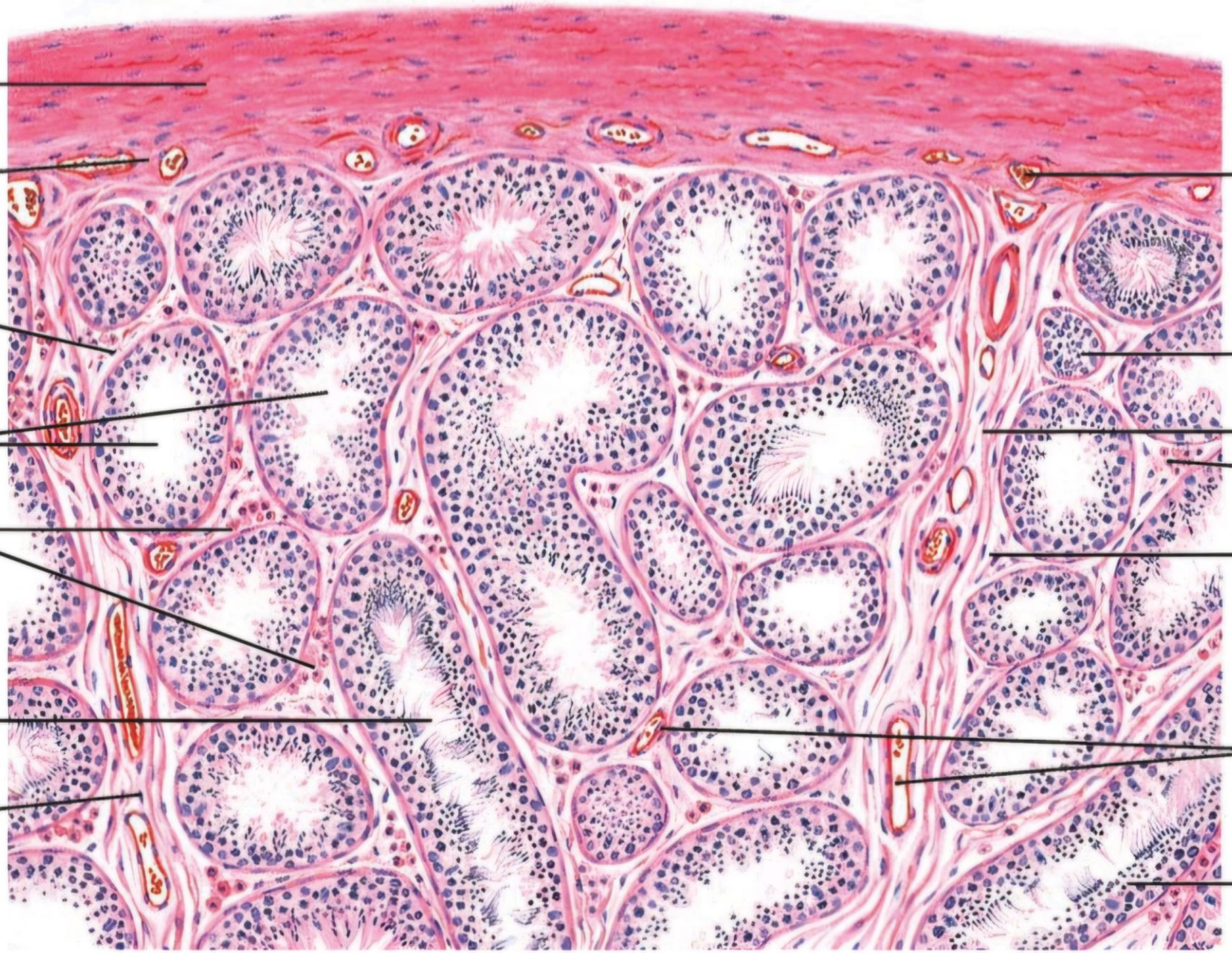


- The male reproductive system consists of external (penis, scrotum, testes) and internal (epididymis, vas deferens, seminal vesicles, prostate) genitalia.

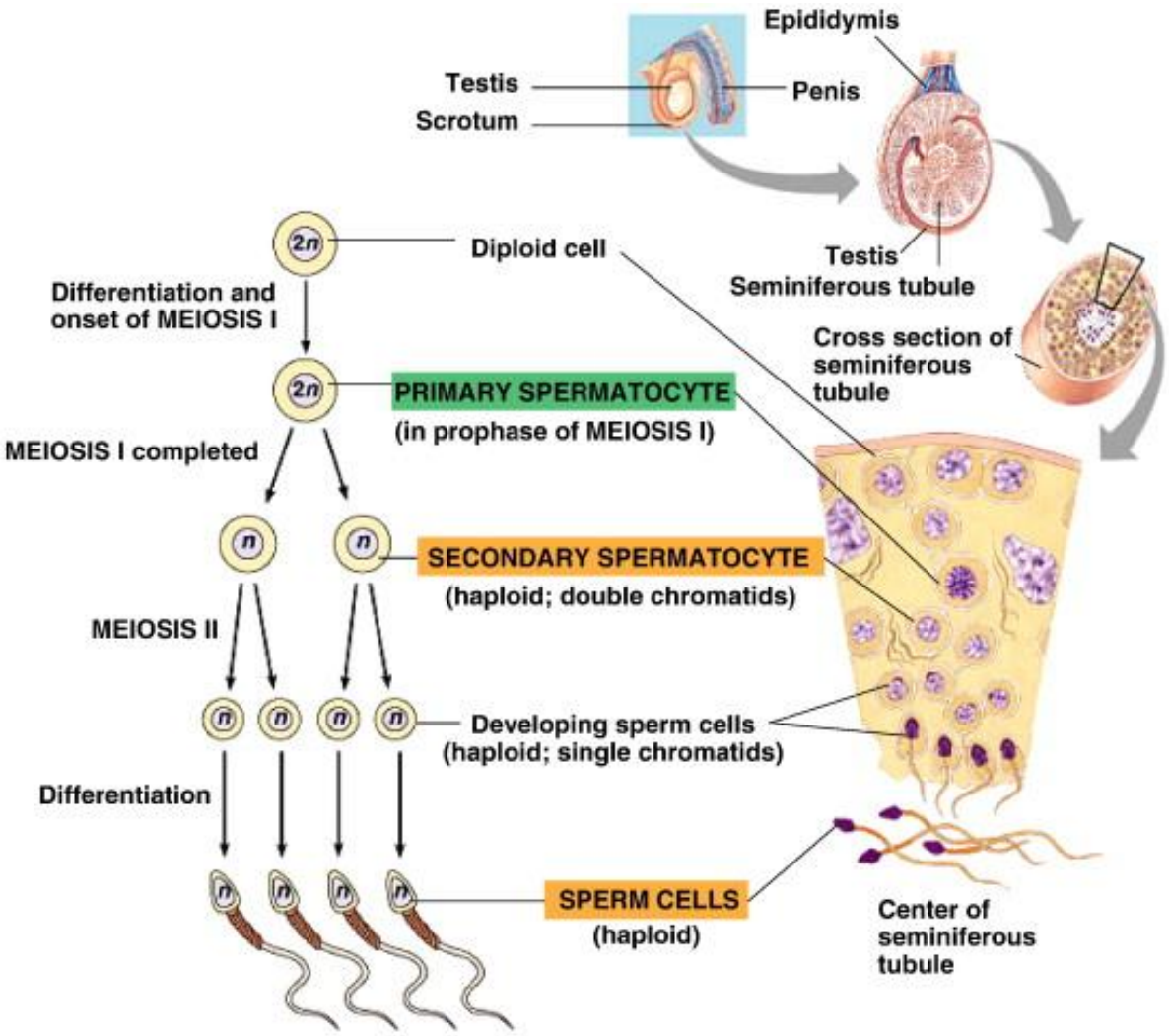


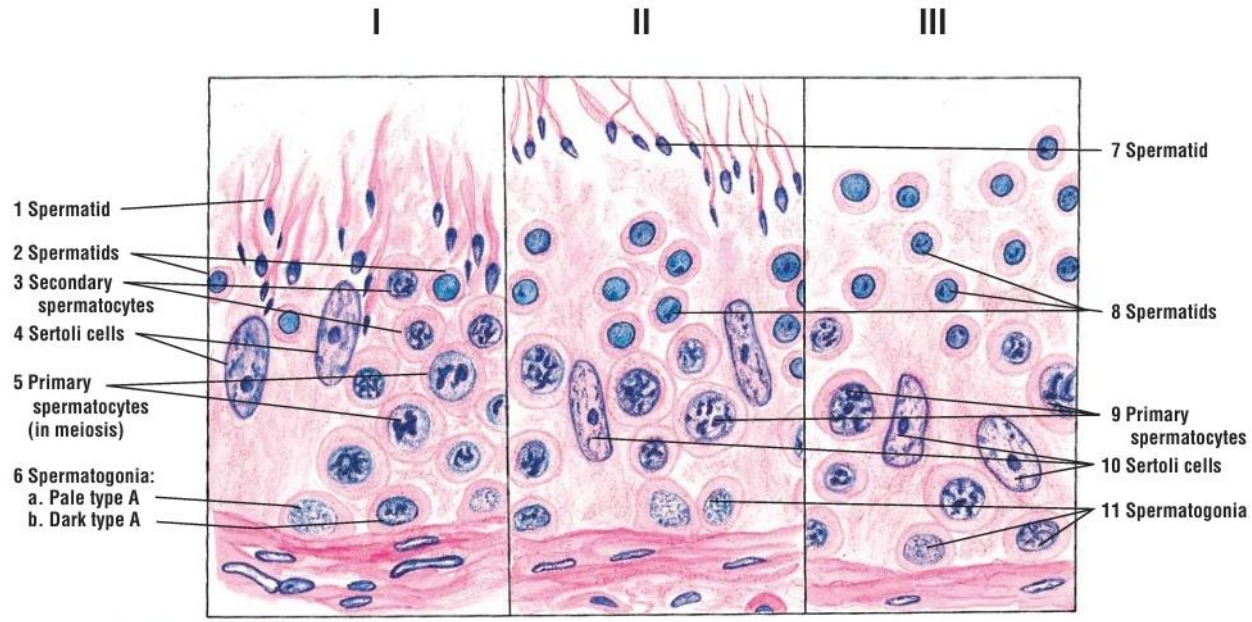


MICROSCOPIC STRUCTURE OF THE TESTIS

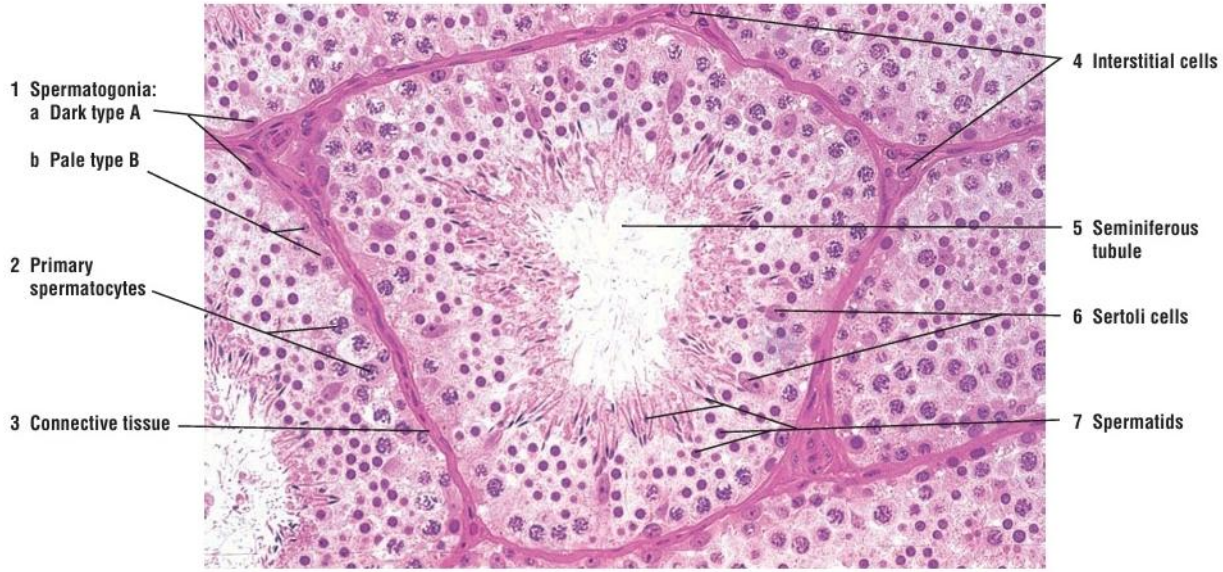


SPERMATOGENESIS

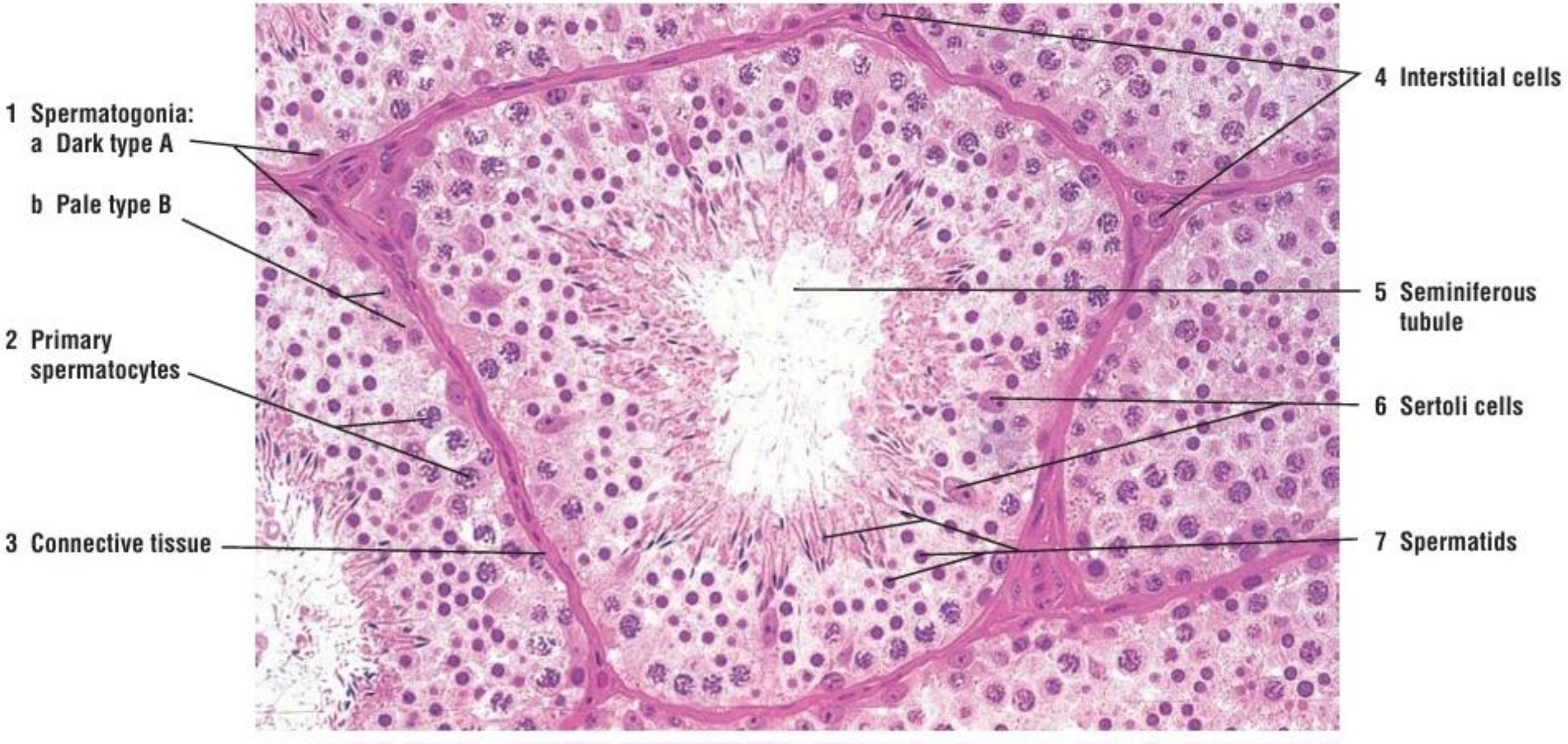




SEMINIFEROUS TUBULE



SEMINIFEROUS TUBULE



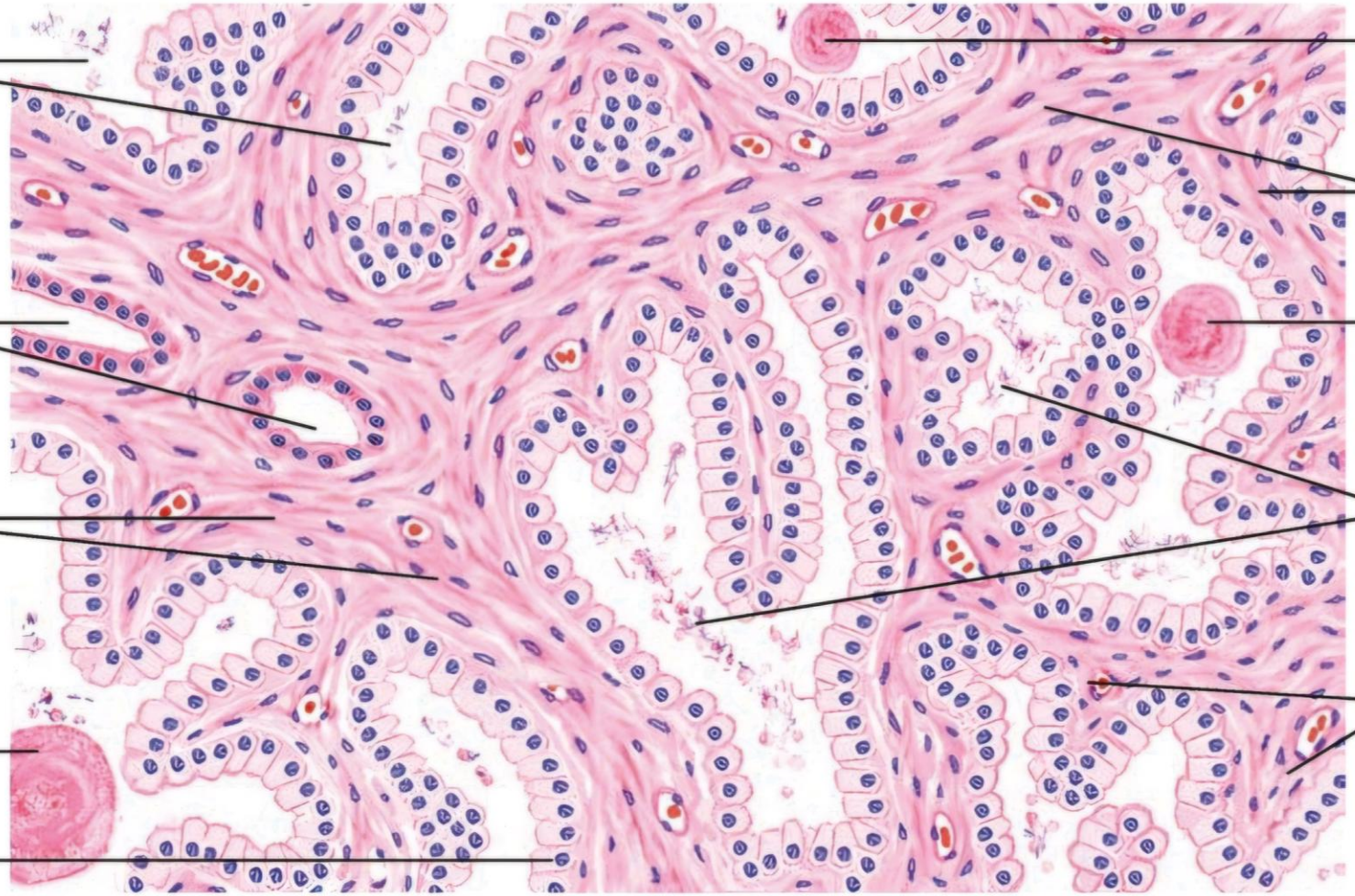
MICROSCOPIC STRUCTURE OF THE TESTIS

- Tunica albuginea – Dense irregular connective tissue capsule; gives septa that divide the testis into lobules.
- Tunica vasculosa – Highly vascular loose connective tissue layer lining the inner surface of tunica albuginea; rich in blood vessels supplying the testis.
- Seminiferous tubules – Highly coiled, circular/oval in cross-section; lined by stratified germinal (seminiferous) epithelium surrounded by interstitial connective tissue.
- Germinal epithelium (lining epithelium) – Complex stratified epithelium composed of spermatogenic cells + Sertoli cells resting on a basement membrane.
- Spermatogonia – Small, round cells with dark nuclei; lie on the basement membrane (basal layer).
- Primary spermatocytes – Large cells with coarse chromatin; located above spermatogonia (most prominent cells).
- Spermatids & spermatozoa – Small, elongated cells near lumen; spermatozoa have long flagella projecting into lumen.
- Sertoli cells – Tall columnar cells, indistinct boundaries, oval/triangular nucleus; extend from basement membrane to lumen; support and form blood-testis barrier.
- Interstitial (Leydig) cells – Found between tubules; large, polygonal cells with eosinophilic cytoplasm; secrete testosterone.
- Rete testis – Channels in mediastinum; lined by simple cuboidal to low columnar epithelium (with microvilli, sometimes cilia).

MICROSCOPIC STRUCTURE OF THE PROSTATE



MICROSCOPIC STRUCTURE OF THE PROSTATE



MICROSCOPIC STRUCTURE OF THE PROSTATE

- The prostate gland is an encapsulated organ situated inferior to the neck of the bladder.
- The thin fibrous capsule, which sends septa into the gland dividing it into lobules.
- The urethra that leaves the bladder and passes through the prostate gland is called the prostatic urethra.
- A transitional epithelium lines the lumen of the crescent-shaped prostatic urethra.
- The prostate is a compound tubuloalveolar gland composed of numerous branching secretory units.
- Its glands are usually lined by simple columnar or pseudostratified, though some areas may show squamous or cuboidal cells.
- It contains abundant fibromuscular stroma, made up of smooth muscle fibers and connective tissue.
- The prostatic acini are irregular in shape and size and are embedded within the stroma.
- The excretory ducts of the prostatic glands are lined by columnar and stains darker near the urethra.
- The gland is divided into peripheral, central, and transitional zones, which are important in disease patterns.
- The lumens of the glands often contain corpora amylacea, which are eosinophilic secretory concretions that may calcify with age.
- The glandular epithelium produces a protein-rich secretion, including enzymes such as prostate-specific antigen (PSA).


THANK YOU!